Lab 3 — Architecture Selection & Justification

Self-Service Coffee Kiosk System

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is organized into Presentation, Business, and Data layers, separating the touchscreen interface, order and payment processing, receipt printing, and menu/pricing data.

Reason 1

Layered Architecture provides clear separation of concerns. The User Interface handles customer interactions, the Business Layer handles ordering and payment operations, and the Data Layer stores menu and pricing information.

Reason 2

The kiosk has a small and well-defined set of functions, so a layered design is simpler to understand and maintain than a microservices architecture with multiple independently deployed services.

Security Advantage

The Payment Service can be isolated within the Business Layer, keeping payment processing separate from the touchscreen interface and reducing direct exposure of payment operations.

Performance Benefit

Separating presentation, business processing, and data access keeps each component focused on its specific operation, allowing requests such as menu retrieval and order processing to be handled efficiently.